

Business Intelligence – Task1

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- Introduction

Football analytics is about using data to understand and improve how teams perform. By collecting and analyzing different types of information, clubs can make better tactical and strategic decisions.

In this report, I will show 10 examples of decisions in football that can be supported by analytics. I will also discuss how analytics can affect the game, and if it can be applied to other sports as well.

- Motivation

In the below table, I will focus on 10 different football analytics decisions. The reason I have selected these decisions are:

- 1- It focuses on various aspects of financial, health, performance, and legal matters.
- 2- Such decisions are real-world challenges. Therefore, analytics can provide valuable solutions
- 3- It demonstrates how analytics can be applied to both on-field and off-field decisions.

Decision	Decision Maker	MGT. Level	Decision Type	Variables	Model/Tool
D-1: Reducing the injury rate during the season	Coach + Doctor + Fitness Coach	Tactical	Semi-Structured	IND: sleep quality, previous injury, training load DEP: Injury rate	Decision Tree
D-2: Extend the stadium capacity	CEO	Strategic	Unstructured	IND: number of attendees, stadium infrastructure, fans experience	Cost-Benefit Analysis

				DEP: stadium capacity utilization	
D-3: Select and price sponsorship packages to maximise expected revenue	CEO + Marketing and Sales Director	Strategic	Semi-Structured	IND: contract duration, sponsorship reputation, the revenue amount DEP: select a sponsorship	Cost-Benefit Analysis
D4: Calculate and monitor the total income and expenses to ensure compliance with Financial Fair Play.	CEO + Accountants	Strategic	Structured	IND: salaries, transfer fees, ticket sales, sponsorship DEP: FFP compliance	Financial forecasting model
D5: How to increase revenue from the club store	CEO + Marketing and Sales Director	Tactical	Semi-Structured	IND: pricing, promotions, online traffic DEP: Club store revenue	Linear Regression
D6: Predict the total tickets sold for the season	CEO + Marketing and Sales Director	Tactical	Structured	IND: number of attendees, tournaments participation, number of tickets sold in the past season DEP: number of tickets sold	Linear Regression
D7: Predict the net profit or loss at the end of the season	CEO + Accountants	Tactical	Structured	IND: number of tickets sold, number of new players, left players DEP: total amount of net profit/loss	Linear Regression
D8: Should a player be included in the starting lineup for the next match?	Coach	Operational	Semi-Structured	IND: player's performance in last match/s, opponent strength/style, DEP: the player will be in the starting lineup or not	Decision Tree

D9: Select the best penalty taker among the team players.	Coach	Operational	Structured	IND: player's success rate of penalties, player's mental status DEP: select a penalty taker	Ranking Models
D10: Decide whether to sell or keep a player in the transfer window.	Coach + CEO	Tactical	Semi-Structured	IND: player's performance in the last season/ matches, player's salary, age and transfer offers. DEP: keep or sell the player	Ranking Models

- **How does analytics influence, positively or negatively, the good of the game?**
 - **Positive impact**

1- Evaluating individual and group performance

Through detailed match records such as goals, passes, tackles, running speed, and distance covered helps coaches identify strengths and weaknesses, and supports better tactical decisions.

2- Reduce injuries and improve fitness

Collecting data from multiple sources, such as heart rate and motion sensors, provides valuable insights into the player's condition and health. This helps reduce injuries and improve fitness by enabling the design of personalized training schedules. This is supported by a previous study, which showed that analysing data using tree decision helps reduce injuries by 12% [\[1\]](#).

3- Analyze the opponent's playing style

Collecting data about the opponent's playing style, including tactics and the strengths and weaknesses of individual players, helps the coach design strategies that exploit these weaknesses.

- **Negative impact**

- 1- **Ethical and Psychological Challenges of Player Monitoring**

Constantly monitoring players can add pressure on them so they feel pressure being always being monitored. A systematic review indicated that using AI in sports raises ethical concerns about fairness, transparency, and privacy, and constant monitoring may increase pressure on players [\[2\]](#).

- 2- **Technological Drawbacks and Loss of Entertainment Value**

Using technology in sport can kill the entertainment. Sometimes, like using VAR can take a long time to get a fair decision, which can lead to reducing the match rhythm and players' fitness during the match.

- 3- **Risk of Over-Reliance on Data**

Overusing digital analytics might reduce the coach's experience role, which may limit creativity on the field.

- **Can football analytics be generalized to other sports? How?**

Yes, football analytics can be generalized to other sports because all types of sports can be analyzed to evaluate players' performance, fitness, and opponents' strengths and weaknesses.

In basketball, for example, analytics can be used to study the points scored by a player, the success and failure of passes and blocks, and the distance covered.

In tennis, analytics can measure the speed and accuracy of serves, assess player fitness, and predict the probability of injury.

In swimming, data such as heart rate, movement speed, and muscle recovery can be monitored to improve training and performance.

Therefore, football analytics can indeed be generalized to other sports, since the underlying concept of collecting and analyzing data to support decisions remains the same [\[3\]](#).

- **Conclusion**

In conclusion, football analytics supports the improvement of performance, revenue generation, and player health. However, it also introduces ethical and technological challenges that must be carefully managed. Overall, data-driven decision-making is becoming indispensable in modern sports management.

References

- [1] D. Manescu, "Big Data Analytics Framework for Decision-Making in Sports Performance Optimization," *International Journal of Computer Applications*, vol. 45, no. 3, pp. 55–70, 2023. [Online]. Available: https://www.researchgate.net/publication/393750697_Big_Data_Analytics_Framework_for_Decision-Making_in_Sports_Performance_Optimization
- [2] J. Smith and T. Brown, "Ethical implications of artificial intelligence in sport: A systematic review," *Journal of Sport and Health Science*, Elsevier, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2095254625000250>
- [3] P. Sundaram and S. Patil, "Sports Analytics: Performance Evaluation and Match Prediction in Different Sports Using Machine Learning," *International Journal of Computer Science and Information Technologies*, vol. 11, no. 5, pp. 45–52, 2024. [Online]. Available: https://www.researchgate.net/publication/377601234_Sports_Analytics_Performance_Evaluation_and_Match_Prediction_in_Different_Sports_Using_Machine_Learning